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OPTIMIZATION OF RESOURCE ALLOCATION AND ROUTING IN POLITICAL CAMPAIGN PROBLEMS

A Thesis by

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Submitted to the Department of Industrial, Systems and Manufacturing Engineering
and the faculty of the Graduate School of
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in partial fulfillment of
the requirements for the degree of
Master of Science

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OPTIMIZATION OF RESOURCE ALLOCATION AND ROUTING IN POLITICAL CAMPAIGN PROBLEMS

The following faculty members have examined the final copy of this thesis for form and content, and recommend that it be accepted in partial fulfillment of the requirement for the degree of Master of Science with a major in Industrial, Systems and Manufacturing Engineering.

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Gokcen Akgun

ABSTRACT

In this thesis, we introduce the problem of a politician who has to allocate his limited budget among meeting and advertising campaigns in the most efficient way so that he can get the maximum number of votes. This problem, optimization of resource allocation and routing on political campaign problems (ORARPCM), assumes that campaigns consist of mass meetings and advertisements that include many types like social media ads, TV ads, party flags or license plate frames. While the politician determines the most effective budget allocation for his campaign, he also selects the cities to run each campaign type and creates the best yielding route for mass meetings according to some predefined distance criteria. In this study, the aim is to observe only the effects of different budget allocations on total vote gain. The proposed ORARPCM can be classified as a multi period, dynamic and time-restricted prize collecting traveling salesman problem (PC-TSP) with budget limitation. Under these assumptions, mathematical model is proposed and numerical results of a Turkey case study is provided. As the solution method, a deterministic dynamic programming model is constructed by using C++.

Results show that under the above-mentioned criteria, deterministic dynamic programming is a very efficient solution method for ORARPCM, as the problem is decomposed based on time periods. Besides the computational time is very reasonable.

Keywords: Campaign Routing problem, Resource allocation, Dynamic programming, Vehicle routing problem

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LIST OF ABBREVIATIONS

ORARPCP	Optimization of Resource Allocation and Routing on Political Campaign Problems
PC-TSP	Prize Collecting Traveling Salesman Problem
TSP	Traveling Salesman Problem
VRP	Vehicle Routing Problem

LIST OF SYMBOLS

₺	Lira (Turkish currency)

CHAPTER 1

INTRODUCTION

A political campaign can be defined as a coordinated endeavor geared towards the goal of influencing maximum number of people to the desired candidate, group or thought. The most prestigious political campaigns are the ones run for general elections, presidential elections, local elections or head of state. In a political campaign, every detail including the color of a tie to be worn in a public speech is deliberately planned towards the ultimate goal. So, staying organized is the first and foremost strategy. Particularly with limited time and budget, every effort has to hit the bull's eye at the very first attempt.

The purpose of this work is to propose a model that answers one indispensable question in politicians' minds right before the outset of the campaign period: How to invest the limited time and budget most efficiently to get the most out of campaigning efforts? We can interpret this problem as how to most effectively use monetary funds to shape voters' minds and how to use the limited time to reach the largest number of people possible. Although in a political campaign, certain vote yields cannot be guaranteed for certain amount of resources, it is our aim to develop a model that maximizes the vote yield by selecting the best allocation between advertising and meeting expenditures. Depending on this allocation, a route to conduct mass meetings will also be proposed. Furthermore, we treat time constraints as another factor to keep an eye on, that is, in a limited campaign period, it is preferable to visit cities with greater vote potential and greater probability of winning. By introducing this mathematical model, we aim to shed further light on increasing the campaign effect based on prior election's results.

In Chapter 4, we implement our model to Turkish general elections which is held in every four years to elect members of parliament to be sent to Grand National Assembly. In Turkey, a proportional representation electoral system, also known as D'Hondt method, is used to elect the 550 members of the parliament according to number of votes coming from 85 electoral districts. Political parties have to go beyond a 10% threshold limit nationwide in order to be able to send a Member of Parliament to National Assembly. According to AKP's (the largest political party in Turkey that came to power in 2002) official website, they allocated a budget of 236,024,601₺ which makes $66,718,855 for 2015 elections according to current exchange rates. Acknowledging the giant amount spent, there is no room for adventure here so, even one flyer's destination must be very well planned in order to gain highest benefits from campaigning efforts. From this point of view, every type of electoral systems require political party leaders to always seek ways to increase the campaign effect and by using the available resources most effectively.

A campaign effect can be measured by the difference between what the polls say before the campaign period starts and the actual outcome after the election is held. This difference is expected to be a function of the consumption of resources of all types including time, money, personal effectiveness, etc. (Day, 2010). But most of the time, same amount of spending does not result in the same outcome. Although these differences may be grounded on a variety of factors, in this study, we focused on the resource allocation part in the campaigns. Therefore, the goal is to find the best investment strategy between campaign types to get the maximum number of people affected from the campaign activities because voters are not passive over the course of campaign.

Voters do not make random decisions for whom to vote. Indeed, they watch, observe and derive conclusions. Their decision processes are affected by many factors including their political leanings, physical environments or the information they gather to make a true assessment for

themselves. The information they collect is largely determined by what the candidates prefer to provide, so campaigns are very important on voters' decision processes as it facilitates the information flow from the candidate to the voters.

According to Popkin (1994) voters use low-information rationality, a term which indicates that voters use their observations from the world around them to make a decision for whom to vote. This implies voters do not use mathematical inequalities to plug in some numbers to see which side overrides, but they use their observations of the issues, candidates and political vows. The information they use is mostly neither complete nor precise, so their decisions can be grounded on a variety of factors which we will talk about later in Chapter 2.1.

Although in the past, politicians could only utilize mass meetings to influence voters, with the emergence of mass communication tools, they obtained the chance to reach millions of voters at the same time (Balci & Bal, 2008). In 2016, internet users spent 118 minutes on average, daily on social media which indicates that traditional media is gradually being taken over by interactive media such as Twitter, Facebook or Periscope etc. So this being considered, it creates new ways for candidates to allocate their resources (Neuendorf, 2016). Barack Obama made use of social media so effectively in 2008 presidential elections that in the following years many candidates showed extensive social media presence in order to experience same benefits (Smith, 2011). While taking advantage of mass communication tools like social media is critical, there is one other factor that is even more critical: individual's exposure to advertisements. (Freedman & Goldstein, 1999) tried to estimate an individual's media exposure to political ads from the survey data by measuring the expose as a function of two variables: the frequency of ad broadcasting and quantity of viewing by the individual. Combining their results with the effect of negative advertising, they argued that negative advertising has no effect of demobilizing voters.

In this thesis, all types of advertisement and every mass meeting are assumed to make a positive contribution to the election outcome.

1.1. Motivation

Political campaigns are imperative in modern democracies. Given its weight in nation's future, it has prompted serious studies from researchers. Many aspects of political campaigns have been worked on so far. These studies can be grouped into two based on their focus area. First group can be considered of studies more about campaigns' qualitative characteristics like communication methods (Neuendorf, 2016; Trent & Friedenberg, 2008), negative campaigning (Lau, Sigelman, & Rovner, 2007; Skaperdas & Grofman, 1995) or campaign impact (Farrell & Schmitt-Beck, 2002). The second group that gives more information on campaigns' quantitative characteristics consists of resource allocation (Bartels, 1985; Colantoni, Levesque, & Ordeshook, 1975; Snyder, 1989) and budgetary effects in political campaigns (Jonathan, 2006).

Despite the broad research done on resource allocation in political campaigns, relatively less work on campaign routing problem was available and no researchers has attempted to tie these two subjects together. This research gap motivated us to combine the resource allocation problem together with the campaign routing problem to propose an elaborated political campaign plan which considers the cities that will be visited and their routes. So, in our work we present a new integrated model that finds the most productive budget allocation with the best yielding route to schedule mass meetings according to cities' current vote potentials and the percentage of vote gain in the previous election from the city and the selection of locations that will be given advertisement. This model can be adopted by other countries to deduce a detailed and effective campaign plan.

1.2. Organization

The outline of this study is structured as follows. In the first chapter, a brief introduction of the political campaign problems and the problem in this study is presented. The importance and uniqueness of the problem is expressed as the motivation of the study in the first chapter. The literature of the political campaign problems is reviewed in the second chapter under different relevant sections such as campaign effect, PC-TSP, resource allocation in political campaigns, advertisement effects in political campaigns. In Chapter 3, the deterministic dynamic programming used as the computational method in this study, is described. The proposed mathematical model is also stated in Chapter 3. In Chapter 4, Turkish case study and network is given along with the computational results. Sensitivity analysis for large Turkish case study presented in Chapter 5. A different approach for the developed model in Chapter 3 is described in Chapter 6. Findings are presented as discussion in Chapter 7. Future research directions and concluding remarks are presented in Chapter 8.

CHAPTER 2

LITERATURE REVIEW

Political campaign problems have been studied for many decades. The proposed mathematical model in this study has strong connections to campaign effect, since everything in a campaign is planned towards to maximize the effect; PC-TSP, since every visit to election regions contributes positively to the vote turn whereas every unvisited region may hurt the potential benefits; resource allocation, since the allotment must be done in the most efficient way; advertisement theory, since it is all about the persuasion of people to support.

Therefore, in this study a literature review is presented for the campaign effect, PC-TSP, resource allocation and advertisement effects in political campaigns respectively.

2.1. Campaign Effects

Political campaigns are very important for the election outcome. If a candidate did not campaign at all, many voters would have been uncertain about the candidate which has been found to have a negative effect on voters. A political campaign is established with a particular goal in mind: to win the election. To achieve this goal, a campaign tries to reach as many people as possible and persuade them to support the goal of the campaign, and hopefully make them contribute actively to the campaign itself. So, the effects of campaigns on the outcome and resource allocation among these campaign types should be thought over and over to squeeze all the benefits. There are several studies in campaigns' effects on the outcomes. Day (2010) constructed a formal model for U.S presidential campaigns that integrates the prior probabilities of candidates in 51 locations and different levels of effectiveness rates they might have. His work is worth

investigating so we will explain a bit. According to the model, swing states and higher electoral vote states are assigned more resources by candidates. Also, candidates' effectiveness is an important factor on deciding whether to increase the resources allocated, because at times of top effectiveness, they might experience higher marginal returns than their challengers. Taking those factors into account, he established a formal model and checked its validity by investigating 1976-2008 elections for qualitative support. He also scrutinized 5 elections' data (up to 2008) to quantitatively support his model. Ultimately the model proved valid and also flexible to address different levels of candidate effectiveness which ends up being a contributor to election outcomes.

According to Brady et al. (2006), a campaign effect can be described by asking the question how campaigns influence things that are relevant to voter's choices and directly voter's choices? So instead of narrowing down the scope to a do campaigns matter question, researchers should figure out when and how campaigns matter. In this perspective politicians like Richard Nixon (Niffenegger, 1988) and some researchers have given more weight to battleground regions than blackout regions (Gimpel, Kaufmann, & Pearson-Merkowitz, 2007). But this approach does not always yield the intended results as in the case of Hillary Clinton. In 2016 presidential elections in USA, one of the reasons for her defeat was her taking for granted the Rust Belt States and extremely investing in three swing states; Ohio, Florida and North Carolina. As a result, she did not only lose in those neglected states but also, she could not win in those three swing states.

Gerber (1998) reevaluated the argument that the challenger's spending is more deterministic than the incumbent's spending and asserted that when the endogeneity is considered, each unit of campaign spending approximately yields the same result for both the incumbent and the rival.

The prevailing argument among political pundits is that campaigns do only minimum to change the voters' view. Early studies in this field done on 1940 US presidential campaign suggest that during the campaign period only 8% of the voters altered their opinions towards the other candidate and over 50% had made their mind before the campaign course started and were not affected by the campaigning (Paul F Lazarsfeld, Berelson, & Gaudet, 1948; Paul F Lazarsfeld, Berelson, & McPhee, 1954). But in his subsequent studies (Paul Felix Lazarsfeld, Berelson, & Gaudet, 1968) proposed that campaigning plays a critical role also in persevering the partisans, solidifying their support and as a result less leakage from the ranks.

Iyengar & Simon (2000) demonstrated that contrary to common sense, campaigns did matter and due to insufficiency of conceptualism and methodology of conventional models, it was believed that campaigns did only little. To overcome those roadblocks, they performed a resonance model that keeps the receiver's inclination and the framework of the message in view and a strategy model that assesses campaign messages and candidates interdependently. Similarly, Bartels (1987) argued that once got out of the range of presidential elections, that is, when no party identification spirit manipulates voter's thinking as in the case of presidential primaries, campaign effectiveness mounts up.

A wide range of studies have been conducted to measure the campaign effect on voter's behavior. Lau & Redlawsk (2006) and Kasper Møller Hansen (2008) developed models to explain how voters are affected by the political campaigns. Lau & Redlawsk (2006) constructed an innovative experimental methodology to help the voters reveal how they were affected by the huge information flow during political campaigns. Four decision strategies that voters unintentionally use were characterized and the factors direct voters to these strategies were analyzed.

Kasper Møller Hansen (2008) reviewed six competing models in depth. The first one is civic engagement which argues that voters' political awareness and willingness to vote escalates during the exposure to the campaign (Freedman et al., 2004). (Craig, Kane, & Gainous, 2005) also suggested that exposure to campaign results in an increase in the difference between the most proficient and the less. The priming studies, the second one, suggests that campaigns guide voters on according to what criteria they will assess the parties, candidates or leaders (Togeby, 2007). The third one is minimal effect model. This model advocates that campaigns do nothing but fortify voters' initial choices because from a campaign people only pick up and learn arguments that support their initial inclinations (Kasper M Hansen & Kosiara-Pedersen, 2012). The fourth is memory based models which put forwards that campaigns influence the voters only as a function of their political sophistication and political inclination. That means voters who have lesser information and stronger predisposition are the ones least affected by the campaigns. Next is the online based models which advocate that voters tend to quickly forget the discourses that affected their final decision but their decision is continuously shaped by the campaign messages over the course of the campaign (Lodge, Steenbergen, & Brau, 1995). The last one is shortcut based models which emphasize the shortcut factors that do not necessarily call for a voter's deep evaluation of the campaign/politician. For example, a leader's charisma (Popkin, 1994) or green light from a trade union or a family member can be taken as shortcut decision factors.

2.2. Resource Allocation in Political Campaigns

Resource allocation is one of the main themes in economics which necessitates consuming of scarce sources in the most efficient way parallel with the organization's strategic goals. In project management, resource allocation forms the most complex part as the sources are deployed in a fragmented manner over the course of the project execution (Schwindt, 2006). These problems

can be solved by computer programs that specifically apply to certain domains to dynamically allocate resources to cost items. As an example, Rathinam et al. (2007) demonstrated a constant factor approximation algorithm to be used in multi-vehicle systems that have non-holonomic constraints.

In political campaign planning, resource allocation problem is a main hurdle. Despite the fact that it has the potential to revert the election outcome, it has not received satisfactory attention from researchers although a few study found in the literature. Brams & Davis (1982) developed a model that aims to optimize the allocation of resources for candidates to use in presidential primaries. One of the most remarkable assumptions they adopted was thinking momentum as of something passable from prior primaries to next ones. They formulated this transfer by simply adding a coefficient into the formula that corresponds to the probability that a voter voted for that candidate in the preceding election. By doing this, they ensured the transfer on the overall sequence of primaries. Brams & Davis (1974) proposed a model that suggests that candidates should allocate their budget among fifty states according to a rule which implies that the number of electoral votes of each state should be raised to 3/2's power to determine budgets for each state. This formulation signifies the fact that states with higher number of electoral votes should be prioritized among others. The validity of results was tested empirically by proving compliance with elections from 1960 through 1972. This model is also pertinent to Colantoni et al. (1975) exponential model on resource allocation since they incorporated a parameter β as an exponential for the magnitude of electoral votes.

Leuthold (1968) defined a political campaign as "the process of acquiring and using the political resources that can secure votes." Defined as doing the necessary in the pathway to electoral victory, Bartels (1985) extended the limits of rationality to incorporate instrumental

(campaign appearances and advertising) and ornamental utilizations (state level funds and personnel) of total resources. He contributed that according to both the Brams and Davis' 3/2 model and Colantoni's multiplicative model, ornamental uses was not treated as main contributors so their contribution was neglected since they are small.

Snyder (1989) investigates the behaviors of two political party under two different goals. These two different goals lead to different attitudes in parties. Their results indicated that in closer races, parties were required to spend higher amounts than in safe districts which was consistent with the previous work done by (Jacobson & Kernell, 1984).

2.3. Advertisement

Political advertisement is the most important instrument in political campaigns to influence voters. Dates back to 1950s, it was first used in the USA then dispersed all around the world. Most of the time, it is created by political consultants and used to form debates, namely political parties or candidates may bring up any issue they want to be discussed in their advertisements and it gives them the opportunity to shape voters' minds. So, advertising in political campaigns is a painless way to manipulate voters' minds which is why oftentimes campaigns are attributed with advertisements in people's eye. Although its effectiveness and suitability to convey a party/candidates' activities and commitments, many countries including United Kingdom and Ireland do not allow paid advertisements on TV or radio doubting that voters will be exposed to advertisements of only those who hold a certain wealth level.

As time evolved so was technology. It was first in 2006 and 2008 elections that young American people utilized technological advances thoroughly to create and exchange the political content among social media users that played a significant key factor in shaping their political decisions (Hesseldahl, MacMillan, & Kharif, 2008).

Goldstein & Freedman (2002) investigated the effects of campaign advertising by scrutinizing 1996 presidential and national election data to see if it really shrank voter turnout. In contrast with demobilizing proponents (S. Ansolabehere, Iyengar, Simon, & Valentino, 1994) and (S. Ansolabehere & Iyengar, 1995) they concluded that both negative and positive advertising stimulates voter turnout with the argument that that opposite results stemmed from lack of data or inaccurate models. Specifically criticizing S. D. Ansolabehere et al. (1999)'s work for being lack of accurate data, they introduced new national ad tracking data of greater scope to measure the ad exposure more precisely in 1996 presidential election.

Roberts & McCombs (1994) investigated the relationship between news agenda setting and campaign advertising. Subjecting 1990 Texas gubernatorial issue agendas of intermedia coverage to content analysis and utilizing cross-lag analysis, they found strong correlation between newspaper, television agenda, and campaign advertising. So, their findings approved political consultants that advertisement shaped the newspaper's and television's coverage of the issues.

Franz et al. (2008) criticized Krasno & Green (2008)'s work of being imprecise arguing that their finding of campaign advertisement did nothing to shape voter's minds was unconvincing for three reasons; firstly they used aggregate data over individually conducted surveys, secondly the misuse of fixed effects model and lastly the generalization of a study built upon only 2000 presidential elections.

2.4. Vehicle Routing Problem

The vehicle routing problem (VRP) is classified as an integer programming and combinatorial optimization problem. For a fleet of vehicles, VRP finds the optimal set of routes to deliver to a set of customers. This problem was first studied by George Dantzig and John Ramser where an algorithmic approach was introduced for the petrol deliveries (Dantzig& Ramser, 1959).

VRP provides a basis for the service of delivery companies by answering the question how to distribute from one or multiple depots by a set of vehicles operated by a crew of drivers who can travel on a given network to deliver customers. So, basically VRP investigates the best set of routes that give the global minimum cost formed by each vehicles' route. While doing this, customer requirements (such as time windows or amount) and operational constraints must be met.

The objective function and the structure of a VRP can be very different depending on the application. In our problem, there is only one depot shown by "capital" and only one vehicle shown by a "politician." Moreover, the vehicle capacity shows the "budget and distance" limitations. A politician visits some of the "cities," defined by "customers" in a classical VRP, which yield the maximum votes. Figure 2.1 illustrates an example of a classical VRP. The objective is to decide on the routes under the budgetary and distance constraints to provide the maximum number of votes and win the election. With this form, our problem can be considered as a classical VRP.

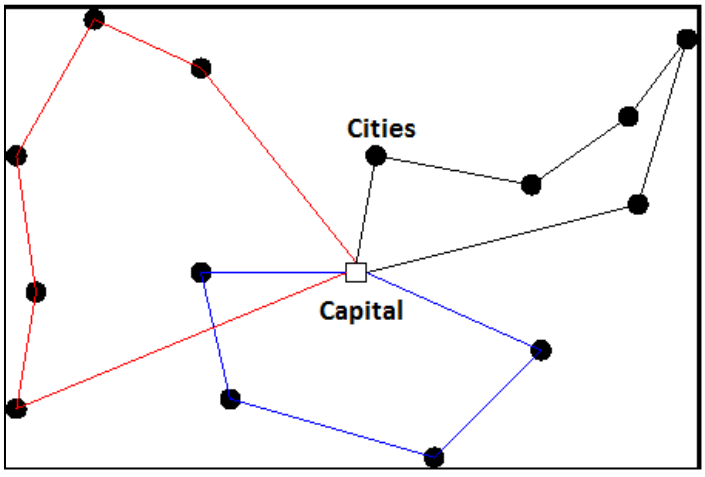


Figure 2.1: A figure illustrating VRP

VRP generalizes the well-known Traveling Salesman Problem (TSP). TSP can be summarized as given a set of customers (cities), the aim is to find the shortest route possible when each customer (city) needs to be visited one and only once and when the loop should be finished at the starting point. There are some types of TSP including PC-TSPs, TSP with time windows, multiple TSPs etc. Our problem is similar to PC-TSP that a politician has to collect the maximum number of prizes (votes) by visiting cities. PC-TSP problems are reviewed in the next section.

2.5. Prize-Collecting Traveling Salesman Problem (PC-TSP)

In our problem, we have a politician who must travel as much as he can in order to collect the maximum number of votes (in the context prize) and minimize the penalty of loss of votes. Additionally, the politician must finish at where he/she initiated the tour and he/she may skip visiting some cities for the sake of campaign because of the budgetary and time restrictions even though there is a penalty initiated with it (Faigl& Hollinger, 2014). But this decision is taken with the consideration of the cities' relative vote yields. In this regard, our problem has very similar constraints with a typical PC-TSP whose basic concept is thoroughly explained by Balas (1989).

In the literature we see a wide range of studies on PC-TSP including very interesting areas of applications like traveling tournament problem where the aim is to optimize sports timetabling (Biajoli & Lorena, 2007). Anderson et al. (2015) attempted to devise an optimization model for the Kidney Paired Donation Program and developed an algorithm from the integer formulation of PC-TSP and found it very effective to utilize. Subramanian, et al. (2010) approached a VRP that pickup and delivery occurring concurrently by using a parallel heuristic introduced into a multi-start algorithm.

Pedro et al. (2013) developed a tabu search algorithm method to overcome this problem. Tabu search algorithm is a metaheuristic method that employs iterative movements to immediate

neighbors in the search of finding a better solution by taking most recent solutions as tabus to block going towards their direction. As a result, this method proved efficient especially with the larger data instead of using more complicated methods as cluster search.

Fischetti & Toth (1988) used additive bounding strategies by computing every bounding procedure and then employing only the highest among all as the new lower bound on the prize to be collected. Next, they applied branch and bound algorithm to find the optimal solution. They also expanded the definition of the problem by incorporating the situation when a factory needs a maximum amount of supply possible by collecting from a set of suppliers and also proposed the binary search algorithm as a possible solution method. Bienstock et al. (1993) developed an approximation algorithm with constant bound relying mostly upon Christofides (1976)'s algorithm and a rounding method for fractional solutions to be relaxed to integers. Chaves & Lorena (2011) tried to attempt the problem by using a hybrid evolutionary algorithm known as Evolutionary Clustering Search.

Bar-Yehuda et al. (2005) worked on a problem of a repairman whose jobs on a list have different locations and special time window constraints. The goal in this study was to find the most beneficial schedule so that it maximizes the profit. They constructed two models: the first one was for jobs that had release times and deadlines but did not need processing times. An O(logN) approximation algorithm was introduced for this case. The second one was for locations that had asymmetric distances each other so a density parameter was introduced to limit the zigzags between job locations.

CHAPTER 3

OPTIMIZATION OF RESOURCE ALLOCATION AND ROUTING ON POLITICAL CAMPAIGN PROBLEMS

3.1. Problem Definition

A political campaign is built on the idea of reaching to the largest number of people possible under time and budgetary constraints and ultimately adding them to the ranks. Most common practices of a campaign are; mass meetings to interact with the people directly and advertisement to keep the voters exposed to campaign discourse regularly. Everything is done through the achievement of a maximized campaign effect based on former election's results.

In this work, we develop a methodology for the most efficient resource allotment between meeting and advertisement expenditures. For this, different possible budget allocations are created to see which allotment yields the best result. Along with the resource allocation, we propose the best yielding route to conduct mass meetings and selection of the cities to give advertisement under time and budgetary limitations. While selecting cities to visit and give advertisement, model prioritizes the ones that has the maximum potential for the campaigner. Here, the combination of the route and resource allocation problem, is how our work is set apart from the past studies in the field.

Also, in this work, a case from Turkey has been investigated. In Turkish case study, although the length of campaign periods may vary, they are restricted by Supreme Electoral Council (YSK). Also, some specific conventions apply, for instance politicians start the campaign from Ankara, the capital, and usually conduct whistle stop meetings at most in two different election regions and then turn back to Ankara in the same day. So, in our model it is assumed that

for each campaign day politicians depart from Ankara, can visit at most two cities and then come back to Ankara. Campaigning days do not have to be consecutive and the main goal is to obtain the maximum number of votes under the above-mentioned constraints.

In this model, we assumed that there is no competition between politicians, that is, politicians are not affected by their competitor's strategies and do not modify their plan of actions. The only aim is maximizing votes by implementing such a strategy that while optimizing budget allocation between advertisement and meeting costs it also provides the best route to follow for mass meeting campaign.

As the solution method, we propose a dynamic programming model which breaks the problem down to sub problems for each time period and solves them iteratively by storing the data from each step and comparing the solution to the previous solutions to find the optimum one.

3.2. Mathematical Model

In this study, a new selective and time-restricted campaign routing problem is defined based on budget limitation. So, the problem may be classified as a single objective, multi-period, dynamic and time-restricted PC-TSP. The optimization model determines the best budget allocation decision for each campaign including mass meetings and all types of advertisements; the selection decision of cities for each campaign according to the budget allocation decisions, and the order of cities for mass meetings at each time period to maximize the campaign effect over the campaign planning period.

The model has the following parameters: the former election's results for the political party as a percentage, the eligible population to vote in each city, costs involved for each type of campaigns and a distance matrix. A number of possible budget allocations are created

systematically. These different budget allocations imply that x% of the total budget should be spent on advertisements and (100-x)% on meetings, where $0 \leq x \leq 100$. The model provides the best budget allocation among the possibilities that maximizes the total number of votes.

Basic Assumptions:

1. The objective is to maximize the number of votes (total reward) that can be collected by the candidate during the campaign period.
2. There is no competition. We just investigate the campaign allocation effects on votes considering the limited budget.
3. Budget allocation scenarios are systematically generated between the mass meetings and the other advertisements.
4. The candidate collects a reward (vote) from each city.
5. The reward function of a given city k on a given day t takes two parameters:
    a. the voter population of city k;
    b. the vote rate of the party in city k in the previous election;
6. During the campaign period, the candidate cannot stay overnight in one of cities. So, he or she must turn back to the campaign headquarters (capital Ankara).
7. Cities have different results from the previous election to be used in this model (previous election's vote gain as percentage). Candidates have a chance to increase their vote gain in any city by campaigning there. In this model, we view campaigning in a city as a production function. Each unit of resource spent guarantees political leader a certain number of votes in turn. So, campaigning in a city does not have a counterproductive effect on results in our model as we only inspect budget effects on campaigns, not every factor affects it. To

understand and determine the campaign strategy in this work, the above mentioned two concepts are very important.

8. The campaign days are known with certainty before the campaign strategy is planned.
9. In each campaign day, the candidate may visit at most two cities under distance and budget constraints.
10. All cities do not need to be visited. According to the budget and distance constraints, the cities that provide more benefits (votes) are visited.
11. Candidate does advertisements in cities that provide more benefit (vote) under limited budget.
12. Costs and allocated budgets for meetings are equal for each city but allocated budgets for advertisement is determined proportional to cities populations.
13. In each campaign day, the candidate may travel at most *Lmax* km before arriving to the second city.
14. In each campaign day, the distance between the cities that will be visited consecutively must be less than L km.
15. A meeting or advertisement can be held only if the allocated budget satisfies the associated costs.

The first mathematical model assumes that each city is given a certain budget for advertisement. The allocated budget can be as a function of the population of the city, equal for each city, etc. Under these assumptions, the mathematical model is constructed and the following are the model notations that are used in this optimization model.

Indexes:

i: campaign type $i = \{1: advertisement, 2: meeting\}$

*k, j:*city (*k*= 1, 2,…, *K)*

t: time $t = \{1, 2, ..., T\}$

s: order of visiting city

Parameters:

v_k = number of votes in city *k*

C = total budget allocated for all campaigns

d_{kj} = distance between city *k* and *j*

L = maximum travel length between two cities in a campaign day

$L_{\max}$ = maximum total travel length in a campaign day

p_k : initial probability of winning election in city *k*

$a_{ni} = n^{th}$ budget allocation rate specified for each campaign *i*

pop_ratio_k = population ratio for each city *k*

$\cos t_{ik}$ = cost of campaign *i* in city *k*

Decision Variables:

c_i = amount of budget allocated for campaign i

y_{ik} = amount of budget allocated for city k for campaign i

eff_{ik} = effective rate of campaign i in city k

$$x_{ks} = \begin{cases} 1, & \text{if city } k \text{ is visited at the } s^{th} \text{ order} \\ 0, & \text{otherwise} \end{cases}$$

$$u_{kt} = \begin{cases} 1, & \text{if city } k \text{ is visited at period } t \\ 0, & \text{otherwise} \end{cases}$$

$$scen_n = \begin{cases} 1, & \text{if scenario } n \text{ is selected during campaign period} \\ 0, & \text{otherwise} \end{cases}$$

$$adv_k = \begin{cases} 1, & \text{if advertisement is done at city k} \\ 0, & \text{otherwise} \end{cases}$$

$$h_{ik} = \begin{cases} 1, & \text{if budget } y_{ik} \text{ satifies the cost of campaigns in city k } (cost_{ik}) \\ 0, & \text{otherwise} \end{cases}$$

Objective Function:

$$\max z_1 = \sum_t \sum_k u_{kt} p_k v_k eff_{2k} + \sum_k adv_k p_k v_k eff_{1k} \qquad (1)$$

Constraints:

$$\sum_n scen_n = 1, \qquad (2)$$

$$c_i = scen_n a_{ni} C, \qquad \forall i, n \qquad (3)$$

$$y_{1k} = c_1 \, pop_ratio_k, \qquad \forall k \qquad (4)$$

$$y_{2k} = c_2 (1/K), \qquad \forall k \qquad (5)$$

$$\sum_t \sum_k u_{kt} h_{2k} y_{2k} \le c_2 \qquad if\,(y_{2k} > \cos t_{2k}) \qquad (6)$$

$$\sum_k adv_k h_{1k} y_{1k} \le c_1 \qquad if\,(y_{1k} > \cos t_{1k}) \qquad (7)$$

$$\sum_k x_{ks} = 1, \qquad \forall s \qquad (8)$$

$$\sum_t u_{kt} \le 1, \qquad \forall k \qquad (9)$$

$$1 \le \sum_k u_{kt} \le 2, \qquad \forall t \qquad (10)$$

$$\sum_s x_{ks} \le 2 \sum_t u_{kt}, \qquad \forall k \qquad (11)$$

$$\sum_s x_{ks} \ge \sum_t u_{kt}, \qquad \forall k \qquad (12)$$

$$x_{k,s-1} + x_{j,s} \le 1, \qquad \forall k, j, s \ (\text{if } s \text{ is even}) \qquad (13)$$

$$\text{if } d_{kj} \ge L$$

$$x_{j,s} \le 1, \qquad \forall j, s \ (\text{if } s \text{ is odd}) \qquad (14)$$

$$\forall t \left(t = \frac{s+1}{2} \right)$$

$$\text{if } d_{0j} \le L_{\max}$$

$$x_{k,s-1} + x_{j,s} \le 1, \qquad \forall k, j, s \ (\text{if } s \text{ is even}) \qquad (15)$$

$$\text{if } d_{kj} \le L$$

$$if\ d_{0k} + d_{kj} \ge L_{\max}$$

$$eff_{ik} = -49.74 + 5.897 \ln(y_{ik}), \forall i, k \qquad (16)$$

$$x_{ks}, u_{kt}, scen_n \in \{0,1\} \qquad \forall k, s, n, t \qquad (17)$$

$$c_i, y_{ik}, eff_{ik} \ge 0 \qquad \forall i, k \qquad (18)$$

Equation (1), the objective function, aims to maximize the campaign effect by selection of cities to have meetings and give advertisements. When the politician visits city *k*, he can increase his winning probability by the effective rate of eff_{2k} and he can increase the winning probability by the effective rate of eff_{1k} when he does advertisement in that city.

Equation (2) implies that only one possible budget allocation is applied during the campaign period and equation (3) shows the budget allocated for each campaign; advertisement and meeting. Equation (4) and (5) shows the advertisement and meeting budget for each city. In addition, equation (6) and (7) ensures that the total cost of meetings and advertisements held in each city should be less than the budgets allocated for each campaign type, mass meetings and advertisements respectively. Because a politician may visit two cities in a day, we divided a day into two slots to show the order of the cities that will be visited. In each slot, only one city must be visited as shown in equation (8). Because of the budget constraint, all cities may not be visited. So, equation (9) represents that a city may be visited or skipped. In each campaign period *t*, a politician may visit at least one and at most two cities as shown in equation (10). Equation (11) and (12) limit the number of visits for city *k*. Under the distance constraints, after the first meeting in a day, another city may not be visited and the first meeting continues in the second slot of the day. The second city visited in each campaign day must be at most *L* km far from the first city visited, and it is represented by equation (13). Equation (14) and (15) show that the total travel length before arriving to the second city that will be visited in the same day must be less than *Lmax*, otherwise only one city can be visited in a day. According to the budgets spent for each campaign in each city, the value of effective rate function changes. Equation (16) represents the effective rates of each campaign *i*. Finally, equation (17) identifies that all decision variables are binary and equation (18) implies that the variables in the mathematical model are non-negative.

3.3. Computational Methods

In this problem, we propose a special type of dynamic programming modeling due to its applicability to problems composed of many interrelated smaller sub problems. In this method, a systematic procedure is provided for finding the overall optimal solution from the combinations of sub decisions.

Dynamic programming starts tackling the problem by first dividing the problem into smaller portions (stages). Then it finds the optimal solution for either the last or the first sub problem and gradually extends the problem stage by stage to solve it in its entirety. As it can be inferred, it is sometimes a bottom up and sometimes a top down approach that builds a later stage optimal on the former stage optimal. So, the idea is simple, save the output from the already solved preceding sub problem and use it as a future reference in the succeeding sub problem. The computation time is dramatically reduced, but since this method records outputs from each step, time savings are done at the expense of extra memory usage. The process of storing the data is called memorization which is a unique feature not utilized in exhaustive enumeration methods. Also, as opposed to linear programming, in dynamic programming language there is no one standard formulation of a problem as one can adjust equations to suit the problem depending on the degree of his/her past exposure to dynamic programming applications.

Although it is a very effective solution method in optimization problems of economics, mathematics, computer science and management science, this method can only be applied in problems that meet some certain criteria. The basic structure of the problem should carry the below mentioned characteristics to be categorized as a dynamic programming problem:

1. The problem should be able to separate into **stages.**

The most remarkable characteristic of a dynamic programming problem is said to be present if that problem can be divided into simpler stages that when sequentially solved, give an overall solution for the entire problem. These one stage simpler optimization problems are solved as ordinary and each solution in the sequence helps to determine the characteristics of the subsequent one stage problem.

Most of the time optimization problems consist of multiple time periods to make sub decisions, so these time periods can be represented by stages in model. There are also some problems that the stages in it cannot be interpreted in terms of time. This type of problems can still be modeled as dynamic programs but it requires further insight and intuition to recognize them.

In our problem, the stages are the time periods *t* which are called campaign planning periods.

2. Actions are taken during the time period.

In our problem, action specifies the budget allocated for each campaign. The actions are systematically generated and shown as follows:

$$a = [a_1, a_2]$$

where a_1 shows the budget allocation rate for advertisement and a_2 shows the budget allocation rate for meeting. In the case study, we determine 8 different possible budget allocations for meeting and advertisement which means eight different selection of cities and their associated routes will be generated and the model will select the allocation that yields the highest votes by comparing the results to each other. As it can be inferred, a is the action to be taken at the beginning of the

campaign period. Hence it is independent from time *t* and this action is applied till the end of the campaign period.

The best action $a^* = [a_1^*, a_2^*]$ is selected according to total votes yielded among different budget allocation options.

3. Each stage has various **states** either finite or infinite.

A finite or infinite number of states are associated with each stage in the problem. The states reveal the information needed to evaluate the current decision's consequences on future actions.

The identification of the states in a problem is the most critical part in dynamic programming design. Having no roadmap for it, it is widely accepted as an art that requires high creativity and refined intuition about the problem. Some properties that drive the selection of states can be ordered as;

i. Each state itself carries enough information to build future decisions upon without the need of knowing how the process got to that state.
ii. The number of states in a stage should not be too large in order to keep the computational effort required at a reasonable level, otherwise it would be prohibitively expensive that it loses its efficiency. This is one drawback of dynamic programming approach; the suitability to real life problems are limited.

In our model, S_t's are the cities that will be visited at time *t*. In compliance with the constraints, system chooses the city that brings the maximum vote gain among alternatives.

4. At each stage *t* in each state S_t, a transition is made.

The action a in our model is defined as the budget allocation rate scenarios for each campaign and is selected systematically. After taking action a and determining the amount to spend in meeting efforts, the model selects the cities that yield the highest vote (according to the ratio of votes gained to the number of total voters in the previous election and the number of people who is eligible to vote in that city in the forthcoming election). At the end of each period the budget shrinks and a transition from S_t to S_{t+1} occurs by selecting the city that reserves the highest benefit with the remaining budget possible. Hence, this transition is first affected by the action a at the beginning. The value of being at the next state at time t+1 is

$$v_{t+1}(\mathrm{S}_{t+1}) = \max_{a \in A} \left[C(S_{t+1}, a) + v_t(S_t) \right];$$

$$S_{t+1} = f_t(S_t, a)$$

where $f_t(.)$ is the transition function of the process and *C(.) is* the immediate contribution from the selection of S_{t+1}.

In the above representation, the flow is from left to right which implies that the aim is to convert the current state to a future state. When a decision is taken, current state is transformed into a future state. So, a dynamic programming problem can be described as a network and each node (state) in this network is evaluated by its immediate contribution to the objective function.

5. It provides a policy prescription for different cases.

Besides proposing the overall optimal solution, dynamic programming also describes alternative ways for each stage in case of not being able to follow the optimal due to an unplanned incident. This feature is called policy prescription and it provides great flexibility for sensitivity analysis.

6. Principle of optimality in dynamic programming problems requires the optimal for the future stages to be independent of optimal of previous stages.

$$v_{t+1}(\mathrm{S}_{t+1}) = \max_{a \in A} \left[C(S_{t+1}, a) + v_t(S_t) \right]$$

As can be seen from our equation above, given a current state, next move cannot be restricted by previous optimal. In other words, knowledge of the previous stage optimal conveys the information needed to evaluate next stage optimal and then all current values are compared in order to select only the best. This property is known as Markovian Property and problems are not eligible for dynamic programming if they lack this property.

7. A solution procedure might be initiated either from the last or from the first stage.

Forward induction and backward induction are both permissible ways. Although we did forward induction in this study, with a few arrangements including the ones below, we would still get the same results if we chose to do backward induction

$$V_t(S_t) = \max_{a \in A} [C(S_t, a) + V_{t+1}(S_{t+1})],$$
$$S_{t+1} = f_t(S_t, a)$$

8. There is a recursive relationship between stages *t* and *t*+1 that helps identifying the optimum policy for stage *t*+1 from the optimal policy in stage *t*.

Value function, $v_{t+1}(S_{t+1})$, defining the value of being in state S_{t+1} is the number of total votes obtained from the cities that are visited and where advertisements are done, is:

$$v_{t+1}(S_{t+1}) = \max_{a \in A} \left[C(S_{t+1}, a) + v_t(S_t) \right], \quad t = 1, \ldots, T$$

where $S_{t+1} = f_t(S_t, a)$ and $C(S_{t+1} \mid S_t, a)$ is the immediate reward function defining the number of votes obtained from state S_{t+1} when transits from S_t under action a at that time t.

Finally, the optimal action a^* is the action that makes the value function maximum. $a*$, which is the budget allocation scenario that gives the maximum vote is selected as final decision. According to budget allocation decision a^*, the cities that advertisements will be done and the mass meetings will be made will be decided on.

Dynamic programming can be categorized into two groups as deterministic dynamic programming and probabilistic dynamic programming. In probabilistic dynamic programming problems, the state at the next stage does not depend solely on the current state and current decision but there exists a probabilistic distribution as to what the next state will be though this distribution is still completely determined by the current state and decision. The other type is deterministic dynamic programming that we choose as the solution approach for our problem.

3.3.1. Deterministic Dynamic Programming

In this section, the implementation of dynamic programming into deterministic problems will be discussed in detail. In deterministic problems, the next state entirely depends on the current state S_t and the policy decision a.

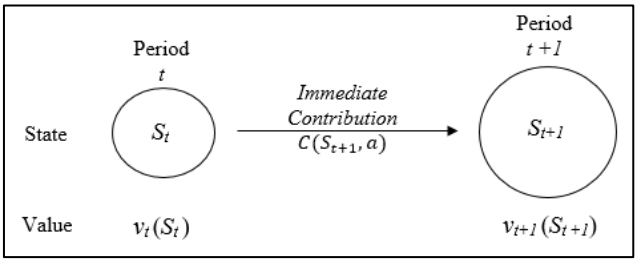


Figure 3.1: Deterministic dynamic modeling framework

So, at stage t, the state is S_t. The policy decision a transfers S_t to S_{t+1} at stage $t+1$. The value function at stage *t*+1 forms from two components: immediate cost (or reward) function of state S_{t+1} at stage *t*+1 and the value function obtained from S_t at stage *t*. To get the value function $v_{t+1}(S_{t+1})$ at stage *t*+1, the value function of $v_t(S_t)$ at stage *t* is previously calculated. Also, some contribution $C(S_{t+1}, a)$ will occur from the action a's side at stage *t*+1. When those two are properly combined, $v_{t+1}(S_{t+1})$ is obtained and the procedure goes one stage forward. Then, the function is optimized regarding to action *a* which provides the maximum vote.

In our model, there is no probabilistic distribution that is involved with the selection of the next state as the next state is entirely determined by the current state and the current policy decision. Therefore, deterministic dynamic programming is an appropriate solution method for our problem.

Deterministic dynamic programs can be recognized by looking at the structure of the objective function. In these problems, mostly, the aim is to minimize a sum of the contributions from individual stages (as distances) or to maximize a sum from individual stages (as in our problem) or to minimize a product. Another way to identify deterministic dynamic programming problems is by looking at the nature of the states in stages. If they can be represented by discrete or continuous state variables, or by a state vector with more than one variable, they can be treated as deterministic dynamic programming problems. Although these differences may seem like a major issue, other than a fluctuation in computational time, they do not make much difference since the underlying structure is mainly the same.

In chapter 4, we provide numerical results for a case study inspired from Turkey.

CHAPTER 4

TURKISH CASE STUDY & RESULTS

4.1. Case Study

In this study, we analyze the proposed model's results for a case study from Turkey. We have gathered Turkish data from various sources. We used data from Supreme Election Council of Turkey for the number of voters in November 1st,2015 general elections in each twenty-one cities that we selected to investigate and AKP's prior election's (June 7th,2015) vote gain as percentage in that cities. The city distance matrix is constructed from the data obtained from Turkish General Directorate of Highway website.

When selecting the 21 cities, we strived for composing a diverse background for both locations wise and population wise to get a sense of real election environment. But while doing this, we ensured to include the three metropolitan cities (Istanbul, Ankara and Izmir) since no politician would take the risk of skipping them. Table 4.1 represents the selected cities used in the model, number of voters in each of those cities, the previous vote gain percentage and the ratio of the voters among the total of voters in the selected cities in this study. The distances between cities are stated in a city distance matrix and represented in Table 4.2.

Table 4.1: Selected Cities, Number of Voters, Previous Vote Gain and Ratio of Voters

Cities	# of Voters	Previous Election Vote Gain%	Ratio of Voters in Each City
Adana	1,484,820	30.4%	4.427%
Ankara	3,745,706	41.17%	11.172%
Antalya	1,589,439	34.9%	4.739%
Bursa	2,046,276	43.9%	6.104%
Diyarbakir	959,078	13.9%	2.859%
Antep	1,139,452	47.3%	3.397%
Konya	1,437,639	65.3%	4.288%
Istanbul	10,318,203	40.9%	30.775%
Van	600,503	16.9%	1.790%
Izmir	3,106,250	26.2%	9.264%
Kayseri	908,971	52.4%	2.711%
Kocaeli	1,241,282	46.34%	3.702%
Elazig	398,411	53.58%	1.188%
Trabzon	561,114	55.5%	1.674%
Tekirdag	673,481	30.1%	2.008%
Samsun	924,381	52.93%	2.757%
Ordu	533,685	53%	1.592%
Mugla	673,642	26.9%	2.009%
Kahramanmaras	700,597	60.7%	2.090%
Sirnak	243,624	9.7%	0.726%
Rize	243,212	66.3%	0.725%
Total # of Votes	33,508,985		

Table 4.2: City Distance Matrix

CITIES	Adana	Ankara	Antalya	Bursa	Diyarbakir	Gaziantep	Konya	Istanbul	Van	Izmir	Kayseri	Kocaeli	Elazig	Trabzon	Tekirdag	Samsun	Ordu	Mugla	Kahramanmaras	Sirnak	Rize
Adana		490	558	837	518	205	356	939	895	900	333	828	486	852	1071	729	718	869	185	711	926
Ankara	490		544	382	910	671	258	453	1232	579	318	342	757	747	585	414	566	620	591	1177	821
Antalya	558	544		537	1076	763	322	716	1453	444	618	605	1044	1236	848	954	1095	311	743	1269	1310
Bursa	837	382	537		1281	313	487	243	1603	322	689	132	1128	1078	375	745	897	541	962	1548	1152
Diyarbakir	518	910	1076	1281		313	874	1363	377	1418	592	1252	153	627	1495	818	729	1387	369	282	701
Gaziantep	205	671	763	1042	313		561	1124	690	1105	353	1013	345	841	1256	725	714	1074	80	506	915
Konya	356	258	322	487	874	561		660	1218	550	304	549	743	922	792	640	781	555	541	1067	996
Istanbul	939	453	716	243	1363	1124	660		1638	561	771	111	1210	1067	132	734	886	780	1044	1630	1141
Van	895	1232	1453	1603	377	690	1218	1638		1762	914	1527	475	716	1770	975	823	1764	746	360	720
Izmir	900	579	444	322	1418	1105	550	561	1762		848	550	1287	1326	505	993	1145	225	1085	1611	1400
Kayseri	333	318	618	689	592	353	304	771	914	848		660	439	618	903	449	484	853	273	859	692
Kocaeli	828	342	605	132	1252	1013	549	111	1527	550	660		1099	956	243	623	775	669	933	1519	1030
Elazig	486	757	1044	1128	153	345	743	1210	475	1287	439	1099		496	1342	665	588	1292	321	435	570
Trabzon	852	747	1236	1078	627	841	922	1067	716	1326	618	956	496		1199	333	181	1367	768	841	74
Tekirdag	1071	585	848	375	1495	1256	792	132	1770	505	903	243	1342	1199		866	1018	121	1176	1762	1273
Samsun	729	414	954	745	818	725	640	734	975	993	449	623	665	333	866		152	1034	645	1094	407
Ordu	718	566	1095	897	729	714	781	886	823	1145	484	775	588	181	1018	152		1186	634	948	255
Mugla	869	620	311	541	1387	1074	555	780	1764	225	853	669	1292	1367	121	1034	1186		1054	1580	1441
Kahramanmaras	185	591	743	962	369	80	541	1044	746	1085	273	933	321	768	1176	645	634	1054		586	842
Sirnak	711	1177	1269	1548	282	506	1067	1630	360	1611	859	1519	435	841	1762	1094	948	1580	586		905
Rize	926	821	1310	1152	701	915	996	1141	720	1400	692	1030	570	74	1273	407	255	1441	842	905	

The summary of input parameters used in the model is given in Table *4.3*. The campaign planning period is determined as ten time periods where each time period consists of two time slots. Two campaign types are considered in the model: Advertisement and Meeting. Eight different budget allocation options are considered in this 21 city case study. The total distance before arriving to the second city cannot exceed 800 km and the distance between two consecutive cities that are visited in the same campaign day can be at most 400 km. Total budget is defined as 31,000,000₺ because AKP approximately spent a total budget of 118,000,000₺ for the November 1st elections and this amount is taken proportionately to the number of cities included in this case study.

Table 4.3: Data Summary

Parameters used in the model	
Number of time periods	10
Time slots in each period	2
Number of campaign types	2 (Advertisement and Meeting)
Total number of cities to be visited	21 cities
Number of budget allocation options	8
Total Budget	31,000,000₺
Max traveling distance before the second city in one day	800 km
Max* distance between two consecutive cities visited in the same day	400 km
Meeting cost per city	1,000,000₺
Advertisement cost per city	1,000,000₺

The proposed model considers eight different budget allocation options presented in Table 4.4.

Table 4.4: Budget Allocation Options for Advertisement and Meeting

Budget Allocation Options		
Options	**Advertisement Budget (%)**	**Meeting Budget (%)**
1	10%	90%
2	20%	80%
3	30%	70%
4	40%	60%
5	60%	40%
6	70%	30%
7	80%	20%
8	90%	10%

To estimate the effect rate, we use some data given in the study of Jacobson (2006). The estimated effects of campaign spending on pre-election knowledge of house candidates are given as in Table 4.5. According to the study, every $5,000 campaign spending increases a challenger's recognition by 2% and so on. We derived an effect rate function regarding to budget invested from these data to use in our model.

Table 4.5: Estimated Effects of Campaign Spending on Pre-Election Knowledge of House Candidates

Budget	Effect rate (%)
5,000	**2**
110,500	**15**
400,000	**26**
800,000	**33**

According to this data, we get the following equation as in Figure *4.1*. *y* denotes the effect rate and *x* is the budget invested.

$$y = 5.897 \ln(x) - 49.74$$

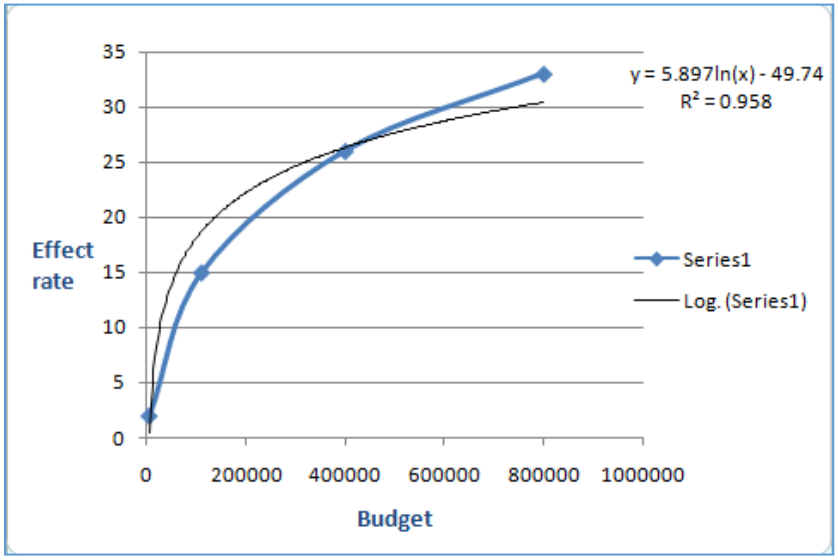


Figure 4.1: The effect rate function

According to the function, the budget and effect rate are highly interrelated with a correlation coefficient of % 95.8.

4.2. Computational Results

For the case study with data in Table 4.3, the following results are obtained with the objective function value of 5,212,410 votes and a total distance of 13,348 km. The best solution has 30% of the budget allocated to advertisements and 70% to meetings. These findings are presented in Table 4.6. Moreover, the politician makes 10 tours totally and visits 16 cities in the campaign planning period which is determined as 10 days (Table 4.7). Additionally, he spends his advertisement budget on Istanbul only (Table 4.8).

Table 4.6: Computational Results

Results		
Total Votes Gained	5,210,000	
Total Traveling Distance	13,348 km	
Selected Budget Allocation	30% Advertisement	70% Meeting

-

Table 4.7: Visited Cities

Visited Cities in Each Period	
T	**Visited Cities**
1	Istanbul, Bursa
2	Konya, Antalya
3	Izmir
4	Kocaeli, Tekirdag
5	Gaziantep,Maras
6	Samsun, Trabzon
7	Kayseri, Adana
8	Ordu
9	Elazig
10	Mugla

Table 4.8: Locations with Advertisement

Base Scenario – Advertisement location
Istanbul

According to the results, a larger part of the budget (70%) is allocated for meeting activities and the rest (30%) is planned to be spent in advertisement activities. The results are consistent with our assumptions: We assumed that each city's advertisement budget is proportional with the population ratio of that city in order to reach as many people as possible in that city but the advertisement costs are equal for each 21 cities. That is, a cost of 1,000,000₺ incurs with the launching of advertisement in each city but more money can be invested with respect to city's population ratio or the vice versa; if the advertisement budget for any city tells to spend less than 1,000,000 ₺, than that city is not given any advertisement since the cost is not afforded. On the other hand, for the meeting part; we assumed equal expenses (1,000,000₺) and equal budget for each city. That is, the available meeting budget for each city is found by dividing the total meeting budget by number of cities and it is not proportional to cities populations. The idea behind it is, because mass meetings take place in large city squares and the rental fee of those squares is assumed fixed regardless of the number of the persons gathered. And since the budget has positive effect on effect rate function, the cities with the higher populations will have higher advertisement effect rates and will provide more votes for the candidate whereas lower populated cities will not be given advertisements since the allocated budget for these cities are lower than the cost of launching advertisement; 1,000,000₺. For example; for Elazig; one of the lightly populated cities, the available budget is found as 109,740₺ by multiplying the 9,300,000₺ (total advertisement budget) by 0.0118 (population ratio). But since the cost of launching advertisement in any city is 1,000,000₺, the model does not select Elazig for advertisement, that is, the initial cost works as a threshold value that helps the campaigner decide which cities are worthy and which cities are not to spend money on for advertisement.

For the mass meetings, the model allocates 70% of the total budget which makes 21,700,000₺ and decides to visit 16 cities. Normally, this amount would be enough to visit all 21 cities, considering the meeting cost of 1,000,000₺ in each city. But the budget is not our only restriction: we also have distance limitations and because of these distance limitations, i.e., some cities are left unvisited. For the mass meetings route; the model puts forward the cities that have the highest vote potential according to previous election's results in that city and the total number of voters. Since two different cities are allowed to be visited in one campaign day, if a secondary city satisfies distance constraints, it can be visited in the second-time slot of the campaign day.

CHAPTER 5

SENSITIVITY ANALYSIS

5.1. Large Network with 21-Time Periods and 50 Cities

In this chapter, a larger case study investigating the impact of budgeting on total vote yield is studied. This case study from Turkey has 50 cities and the campaign period is increased to 21-time periods. Table 5.1 presents the data for the 50-city case study.

In this network, the budget is set at 75,000,000₺. The reason behind this adjustment is, the number of cities used in the network. Turkey is a country of 81 cities but only 50 of those are used in the network and the budget AKP used for November 1st, 2015 elections is around 118,000,000₺. When the budget is taken proportionally to the number of cities, the new budget for 50 cities turns out to be slightly below 75,000,000₺. That is why 75,000,000₺ is a reasonable amount for our larger case study.

Table 5.1: Large Network with 50 Cities and 21-Time Periods, Number of Voters, Previous Vote Gain and Ratio of Voters

No	Cities	# of Voters	Previous Election Vote Gain%	Ratio of Voters in Each City		No	Cities	# of Voters	Previous Election Vote Gain%	Ratio of Voters in Each City
1	ADANA	1,484,820	30.40%	3.156%		26	KIRKLARELİ	265,061	23.70%	0.563%
2	ADIYAMAN	381,687	58.00%	0.811%		27	KOCAELİ	1,241,282	46.34%	2.638%
3	ANKARA	3,745,706	41.17%	7.961%		28	KONYA	1,437,639	65.37%	3.056%
4	ANTALYA	1,589,439	34.96%	3.378%		29	KÜTAHYA	419,508	54.49%	0.892%
5	BALIKESİR	910,091	39.12%	1.934%		30	MALATYA	534,774	58.42%	1.137%
6	BİNGÖL	171,258	47.01%	0.364%		31	MANİSA	1,010,966	37.13%	2.149%
7	BURSA	2,046,276	43.96%	4.349%		32	MARAS	700,597	60.79%	1.489%
8	ÇORUM	385,753	54.28%	0.820%		33	MARDİN	445,455	19.91%	0.947%
9	DENİZLİ	718,251	39.78%	1.527%		34	MUĞLA	673,642	26.93%	1.432%
10	DİYARBAKIR	959,078	13.94%	2.038%		35	NİĞDE	229,962	47.90%	0.489%
11	EDİRNE	304,729	24.77%	0.648%		36	ORDU	533,685	53.00%	1.134%
12	ELAZIĞ	398,411	53.58%	0.847%		37	RİZE	243,212	66.38%	0.517%
13	ERZİNCAN	152,578	49.22%	0.324%		38	SAKARYA	668,777	56.50%	1.421%
14	ERZURUM	483,561	51.97%	1.028%		39	SAMSUN	924,381	52.93%	1.965%
15	ESKİŞEHİR	616,751	36.24%	1.311%		40	SİNOP	154,707	47.54%	0.329%
16	GAZİANTEP	1,139,452	47.31%	2.422%		41	SİVAS	435,685	57.48%	0.926%
17	GİRESUN	320,720	53.46%	0.682%		42	TEKİRDAĞ	673,481	30.10%	1.431%
18	HAKKARİ	157,379	9.71%	0.334%		43	TRABZON	561,114	55.50%	1.193%
19	HATAY	992,286	38.06%	2.109%		44	ŞANLIURFA	977,938	46.78%	2.078%
20	MERSİN	1,218,738	26.42%	2.590%		45	VAN	600,503	16.93%	1.276%
21	İSTANBUL	10,318,203	40.90%	21.930%		46	YOZGAT	295,666	58.12%	0.628%
22	İZMİR	3,106,250	26.21%	6.602%		47	ZONGULDAK	446,210	38.26%	0.948%
23	KARS	182,836	27.17%	0.389%		48	KIRIKKALE	193,688	50.60%	0.412%
24	KASTAMONU	277,244	48.72%	0.589%		49	ŞIRNAK	243,624	9.72%	0.518%
25	KAYSERİ	908,971	52.45%	1.932%		50	YALOVA	168,185	39.73%	0.357%
							TOTAL	**14,079,742**		

5.2. Computational Results for 50 Cities

5.2.1. Base Scenario

The results shown in the table below indicates that, total number of votes to be gained increases as the campaign period, number of cities to run the campaign and the budget increases. The model optimally allocates 30% of the budget to advertisement and 70% to meetings.

Model decides to visit 40 cities and leaves 10 cities unvisited. Among those 10 cities, although Bingol meets the distance criteria and has a high percentage of vote yield from the previous election, it is not selected due to its low population. Another interesting result is, model decided not to have a mass meeting in Diyarbakir, a city widely accepted as a very important eastern city in Turkey, because of the low relative vote yield. This situation stems from the fact that this model does not take any significancy into account other than the numerical data.

Istanbul, Ankara and Bursa are the three cities selected for advertisement campaign. This is because the allocated advertisement budget plays a threshold role to decide whether to give advertisements in cities. City by city advertisement budget is determined proportionally to cities' populations, but costs are fixed. Therefore, if the planned budget is not sufficient for the cost, that city is not given any advertisement. This situation will be reconsidered with a different approach in the next chapters.

Table 5.2: Base Scenario with 50 Cities

Total Votes Gained	7,930,950	
Total Traveling Distance	24,962 km	
Optimal Budget Allocation	30% Advertisement	70% Meeting
# of Cities to Run Each Campaign	Advertisement in 3 cities	Meeting in 40 cities

5.2.2. Gradual Advertisement and Meeting Costs Depending on City's Populations

In the original model presented above, it was assumed that all cities had the same costs for advertisements and mass meetings. But, holding mass meetings in both Istanbul (the biggest metropolitan) and Adiyaman (one of the smallest cities) cannot be the same. Bigger cities require bigger arenas, more party flags, an effective transportation, more complicated audiovisual systems and ultimately more budget. Similar arguments can be carried out for advertisement, too. There lies a huge difference between the number of billboards in Istanbul and Adiyaman. In order to come over this obstacle, a gradually increasing cost approach is implemented in the model according to cities' populations. Table 5.3 shows the classification that was used in this scenario. According to the table, Istanbul, whose population ratio is given 0.21 in Table 5.1, is assigned a cost of 2,000,000₺ for both meeting and advertisement.

Table 5.3: Gradual Campaign Costs

Meeting & Advertisement Costs	
Population ratio ≥ 0.2	2,000,000₺
0.019 ≤ Population ratio < 0.2	1,000,000₺
0.01 ≤ Population ratio < 0.019	750,000₺
Population ratio < 0.01	500,000₺

In this scenario, gradual costs depending on the population is introduced with a total budget of 75,000,000₺. With gradual cost scenario, the optimum budget allocation is 90% for advertisement and 10% for the mass meetings. Model has obviously decided to invest more in advertisement campaign because of the fact that each city was shared the same amount for meetings from the budget, but the costs varied. According to this scenario, in order to conduct a mass meeting even in the lowest cost city, 500,000₺, a total meeting budget of 25,000,000₺ is

required. But even when 25,000,000₺ is allocated for mass meetings, only 20 cities could be given mass meetings because of their lower than 1 % population ratios. That is, from a budget of 25,000,000₺, only 10,000,000₺ could be spent. Because of the inconsistency between cities' meeting budgets and meeting costs, model decides to utilize the money in advertisement campaigns.

Since there is no mass meetings proposed in this scenario, the only vote gain comes from the advertisement therefore the total number of votes gained in this scenario is significantly less than the other scenarios.

Table 5.4: Gradual Campaign Cost Scenario Results (Total Budget: 75,000,000₺)

Total Votes Gained	6,663,480	
Total Traveling Distance	0 km	
Optimal Budget Allocation	90% Advertisement	10% Meeting
# of Cities to Run Each Campaign	No Meeting	Advertisement in 34 cities

Since, the model decided not to hold any mass meetings at 75,000,000₺, the total budget is increased to 85,000,000₺ to see the impact of this increase on the results. According to this scenario 60% of the budget which is 51,000,000₺ is assigned to mass meetings (Table 5.5). One remarkable result is the following: model decides not to conduct a mass meeting in Istanbul because the allocated mass meeting budget for each city is 1,020,000₺. Since different meeting costs are used in this model, the cost of a mass meeting in a city of Istanbul's size is defined as

2,000,000₺. Hence, according to this model a mass meeting in Istanbul is skipped although it is not reasonable to risk the biggest city.

Table 5.5: Gradual Campaign Cost Scenario Results (Total Budget: 85,000,000₺)

Total Votes Gained	7,650,820	
Total Traveling Distance	25,229 km	
Optimal Budget Allocation	40% Advertisement	60% Meeting
# of Cities to Run Each Campaign	Meetings in 39 cities	Advertisement in 7 cities

This result stems from conflict between the assumptions of population dependent meeting costs but fixed meeting budgets for each city regardless of the size. To overcome this problem, proportional meeting budgets will be added to the model along with the gradual meeting costs.

5.2.3. Gradual Meeting and Advertisement Costs & Proportional Meeting Budgets

In this section, the effect of proportional meeting budget with gradual costs will be investigated by comparing the results to the results of the model with the fixed meting budget. Total budget is defined as 75,000,000₺. Results for this scenario are presented in Table 5.6.

In this model 20% of the budget is allocated for meetings and the rest is used in advertisements. It is decided to have a mass meeting only in Istanbul and is facilitated by proportional meeting costs. Because the total meeting budget would be 15,000,000₺ and the meeting budget for Istanbul would be 3,150,000₺ (meeting budget*population ratio of Istanbul) which is above the cost of 2,000,000₺.

Table 5.6: Gradual Meeting and Advertisement Costs & Proportional Meeting Budgets

Total Votes Gained	7,789,930	
Total Traveling Distance	906 km	
Optimal Budget Allocation	80% Advertisement	20% Meeting
# of Cities to Run Each Campaign	Meeting in 1 city	Advertisement in 29 cities

Although in this case the locations with advertisement are five less than the model in section 5.2.2, the objective function value is 17% is higher because Istanbul has a very big potential for the party.

Next, the same model will be applied for a total budget of 95,000,000₺ to observe the selection of mass meeting given cities since with 75,000,000₺ of budget, only one city was selected.

Table 5.7: Gradual Meeting and Advertisement Costs & Proportional Meeting Budgets (Total Budget: 95,000,000₺)

Total Votes Gained	9,252,230	
Total Traveling Distance	5,024 km	
Optimal Budget Allocation	60% Advertisement	40% Meeting
# of Cities to Run Each Campaign	Meeting in 7 cities	Advertisement in 26 cities

Total number of votes has increased by 18% because of the budget increase. According to these results in Table 5.7, meetings are planned in 7 cities and advertisements in 26 cities. It is

clear that locations with advertisements far outweigh locations with meetings. These results stem from the fact that while selecting the locations to run the advertisement campaign, model directly prioritizes the ones that have the highest potentials and a city is said to have a high potential if its value of previous election's vote gain percentage times number of available voters is high. But in the meetings part, since the model both selects and puts the cities in order simultaneously, the cities that take place near the top may not be exactly in order according to their vote potentials. So in this scenario, since each campaign has the same criteria, model decides to invest more in advertisement campaign.

Van is not selected as an advertised location anymore because in the previous model, it received a budget of 760,000₺ which was sufficient for the cost of 750,000₺; but in this scenario the advertisement budget for Van is 727,000₺ which is below the cost.

5.2.4. Gradual Meeting and Advertisement Costs &Proportional Meeting Budgets & Partial Advertisement

Allowing partial advertisement in cities even if the politician cannot afford the total advertisement cost is the analysis done in this section. In the previous scenarios, if the city by city allocated advertisement budget is even 1₺ less than the advertisement cost, politician was not able to place any ad in that city. So even when the politician cannot afford the total cost of advertisement in a city, in this scenario, it is possible to have partial advertisement by spending only whatever the allocated budget satisfies. Therefore, more realistic results are expected from this model because meeting the cost is not a condition anymore to run the advertisement campaign. Table 5.8 shows the results of this scenario.

Table 5.8: Gradual Meeting and Advertisement Costs & Proportional Meeting Budgets &Partial Advertisement

Total Votes Gained	10,799,700	
Total Traveling Distance	16,446 km	
Optimal Budget Allocation	20% Advertisement	80% Meeting
# of Cities to Run Each Campaign	Meetings in 25 cities	Advertisement in 49 cities

By facilitating the partial advertisement, model decided to spend 20% of the total budget in advertisements in 49 cities. In the previous scenario (Table 5.6), because of the cities' populations' related budgetary constraints, although 80% of the budget was allocated for advertisements, only 29 cities were given advertisement. A large part of the budget was not spent due to the shortcoming of cities' assigned advertisement budgets. But in this scenario, even though advertisement took a lower share of 20% from the budget, it was spent completely in 49 cities and since more budget (%80) remained, model selected 25 cities to carry out mass meetings. As a result, the objective function's value has increased significantly.

5.2.5. Gradual Meeting and Advertisement Costs & Proportional Meeting Budgets & Partial Advertisement & Weight

In the original model, the objective function is composed of two sections; advertisement and meeting sections. In both sections, an effect rate function was utilized to estimate the effects (vote yield) of certain amounts of resources spent. But, according to that model, both campaign types yield the same number of votes for the same amount of budget. So, in order to see the impact of each campaign type separately, meeting's effect rate function was assigned a weight of 1.43 in

the objective function in this scenario. This number is found in a study made by Uslu (2009). In her research, she conducted a survey that voters in Turkey were asked how much they had been affected by different campaign types. The results enabled the comparison between advertisement and meeting effects. It was revealed that mass meetings were nearly 1.43 times more powerful than advertisements in a pre-election environment. Therefore, the previous model was run under this weight to see the results.

According to the results shown in Table 5.9, model decided to invest remarkably more in meeting campaigns. This is an expected result since for every unit of resource spent, meetings yield .43 times more vote than advertisements. Moreover, the total distance to be taken during the campaign period has gone up to 20,119 km from 16,446 km due to the increase in meeting budget. Since meetings were made more effective than before, the total number of votes has increased to 18.8 million from 10.8 million.

Table 5.9: Gradual Meeting and Advertisement Costs &Proportional Meeting Budgets& Partial Advertisement& Weight.

Total Votes Gained	18,772,800	
Total Traveling Distance	20,119 km	
Optimal Budget Allocation	10% Advertisement	90% Meeting
# of Cities to Run Each Campaign	Meetings in 31 cities	Advertisement in 49 cities

From this point on, the distance constraints and different budget levels will be analyzed.

To analyze the effects of distance constraints, new distance limits are set in the model. In the base scenario, the distance between two consecutive cities to visit in the same day (L) is not

allowed to exceed 400 km and the politician can travel at most 800 km to visit the first and the second city (*Lmax*) that will be visited in the same day. For example, if the politician goes to Adana as the first city which is 489 km far from Ankara, he/she cannot go to Kayseri as the second city which is 332 km far from Adana because the total distance that needs to be made in the same day before Kayseri is 821 km and which is more than 800 km. Since the emergence of modern transportation methods enabled shorter times for longer distances, *L* and *Lmax* is set at 500km and 1000km, respectively. The results with relaxed distance limits are presented in Table 5.10.

Table 5.10: Gradual Meeting and Advertisement Costs &Proportional Meeting Budgets& Partial Advertisement& Weight (*L*=500 km, *Lmax*=1,000 km).

Total Votes Gained	18,954,700	
Total Traveling Distance	23,444 km	
Optimal Budget Allocation	10% Advertisement	90% Meeting
# of Cities to Run Each Campaign	Meetings in 32 cities	Advertisement in 49 cities

Results indicate that total traveling distance has increased due to an increase in *L* and *Lmax*. Diyarbakir and Mardin are newly added cities to the route and some cities are removed, so the order has changed. A small increase in the total number of votes gained is observed which is because Diyarbakir and Mardin are not very high potential cities for AKP.

5.2.6. Effects of Different Budget Levels

The effect of different budget levels is investigated for the proposed model. In this scenario, gradual meeting and advertisement costs, proportional meeting and advertisement budgets, partial advertisements and weighted approach is considered to analyze the impact of different budget

levels on the final results. Table 5.11 represents the total number of gained votes, trip distance and budget allocation for each budget levels.

Table 5.11: Results from Budget Analysis

Budgets	Total Votes Gained	Total Traveling Distance	Selected Budget Allocation	
			Advertisement	Meeting
60,000,000₺	16,865,800	16,836 km	10%	90%
75,000,000₺	18,954,700	23,444 km	10%	90%
90,000,000₺	20,033,900	23,444 km	20%	80%
105,000,000₺	21,025,300	25,239 km	30%	70%

From Table 5.11, it can be concluded that initially mass meetings have the major impact on the results but vote gain from mass meetings reached to a peak level at 75,000,000₺ budget. At 60,000,000₺ total budget, model allocated 54,000,000₺ for meetings which is the highest amount from possible budget allocations. At 75,000,000₺ total budget, 67,500,000₺ is allocated for meetings. At 90,000,000₺, meeting's budget share fell down to 80% but it is still 72,000,000₺. At 105,000,000₺, the meeting budget is at 70% which makes 73,500,000₺. So, one takeaway from this table is, no matter how high the total budget is, model keeps the meeting's budget around 70,000,000₺. This situation may result either from the distance constraints that prevent the model investing more in meeting or cities' really low vote yields. As a result, no matter how high the total budget is, the total traveling distance does not increase significantly. Budget increase from 75,000,000₺ to 90,000,000₺ supports this argument as the traveling distance remained the same even though the budget increased by 15,000,000₺. So, after 75,000,000₺ total budget, model decides to invest more in advertisement campaign as it still contributes to total number of votes. Figure 5.1 represents these relations in a graph.

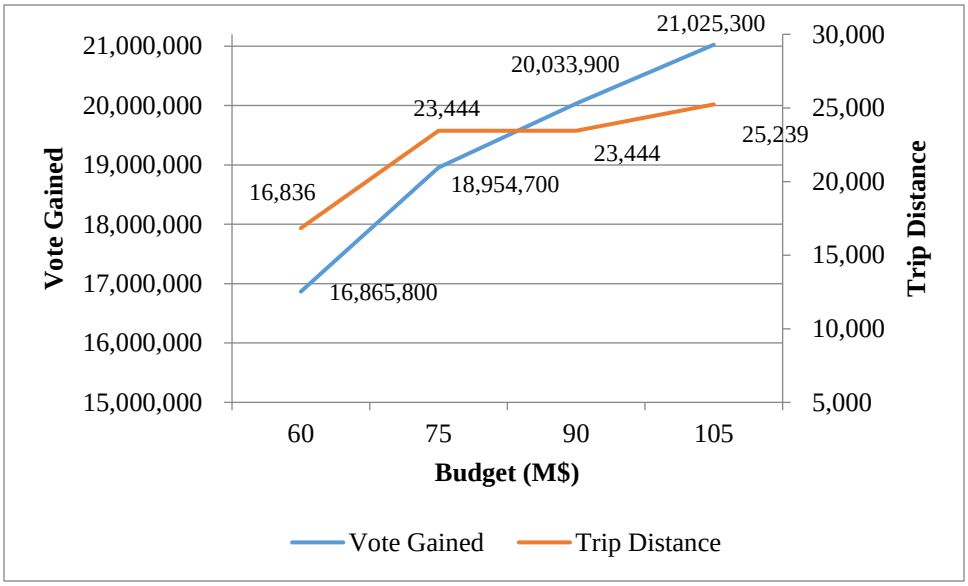


Figure 5.1: Results for different budget levels.

Total number of votes gained shows an increase in the investigated interval. At first, this increase was sharp but after reaching a total number of votes of 18,954,700, the graph showed a slower increasing trend since the number of votes can be gained through meetings has reached almost a peak level and the main contribution came from advertisements only.

In chapter 6, sensitivity analysis is continued with a different approach.

CHAPTER 6

A DIFFERENT STRATEGY IN ALLOCATING THE BUDGET

In the previous chapter, each city had a predetermined advertisement and meeting budget and these budgets were allocated proportionally to cities' population, i.e., every city was shared a budget for meeting and advertisement from the total meeting and total advertisement budgets even though that city is not run any type of the campaigns. It is apparent that this budget allocation strategy may not utilize the whole available budget although the additional spending could increase the overall votes gained. In this chapter, we will let the model to allocate the budget to cities which will surely increase the overall vote gained using a central budget, which will result in a mathematical model that has knapsack type side constraints.

Knapsack problem is one of the basic problems in combinatorial optimization that optimizes the combination of items to include in a collection so that the value maximizes. In this chapter, a budget allocation strategy is developed to adapt the same argument that knapsack problem proposes. The city by city allocated budget will be removed to avoid those never spent gaps in the total budget. Instead, the total advertisement and meeting budget will be distributed only among cities that surely receive a campaign type.

6.1. Mathematical Model

A knapsack problem influenced mathematical model is developed in this chapter as an extension of the model developed in Chapter 3. The proposed model aims to avoid the unused budgets given to the cities that are not selected to run the campaign.

Indexes:

i: campaign type $i = \{1: advertisement, 2: meeting\}$

k, j: city (k= 1, 2,…, K)

t: time $t = \{1, 2, ..., T\}$

s: order of visiting city

Parameters:

v_k = number of votes in city k

C = total budget allocated for all campaigns

d_{kj} = distance between city k and j

L = maximum travel length between two cities in a campaign day

$L_{\max}$ = maximum total travel length in a campaign day

p_k : initial probability of winning election in city k

$a_{ni} = n^{th}$ budget allocation rate specified for each campaign i

pop_ratio_k = population ratio for each city k

$\cos t_{ik}$ = cost of campaign i in city k

Decision Variables:

c_i = amount of budget allocated for campaign i

eff_{ik} = effective rate of campaign i in city k

$$x_{ks} = \begin{cases} 1, & \text{if city } k \text{ is visited at the } s^{th} \text{ order} \\ 0, & \text{otherwise} \end{cases}$$

$$u_{kt} = \begin{cases} 1, & \text{if city } k \text{ is visited at period } t \\ 0, & \text{otherwise} \end{cases}$$

$$scen_n = \begin{cases} 1, & \text{if scenario } n \text{ is selected during campaign period} \\ 0, & \text{otherwise} \end{cases}$$

$$adv_k = \begin{cases} 1, & \text{if advertisement is done at city k} \\ 0, & \text{otherwise} \end{cases}$$

Objective Function:

$$\max z_1 = \sum_t \sum_k u_{kt} p_k v_k eff_{2k} + \sum_k adv_k p_k v_k eff_{1k} \qquad (1)$$

Constraints:

$$\sum_n scen_n = 1, \qquad (2)$$

$$c_i = scen_n a_{ni} C, \qquad \forall i, n \qquad (3)$$

$$\sum_t \sum_k u_{kt} \cos t_{2k} \le c_2 \qquad (4)$$

$$\sum_k adv_k \cos t_{1k} \le c_1 \qquad (5)$$

$$\sum_k x_{ks} = 1, \qquad \forall s \qquad (6)$$

$$\sum_t u_{kt} \le 1, \qquad \forall k \qquad (7)$$

$$1 \le \sum_k u_{kt} \le 2, \qquad \forall t \qquad (8)$$

$$\sum_s x_{ks} \le 2 \sum_t u_{kt}, \qquad \forall k \qquad (9)$$

$$\sum_s x_{ks} \ge \sum_t u_{kt}, \qquad \forall k \qquad (10)$$

$$x_{k,s-1} + x_{j,s} \le 1, \qquad \forall k, j, s \ (\text{if } s \text{ is even}) \qquad (11)$$

$$\text{if } d_{kj} \ge L$$

$$x_{j,s} \le 1, \qquad \forall j, s \ (\text{if } s \text{ is odd}) \qquad (12)$$

$$\forall t \ \left(t = \frac{s+1}{2} \right)$$

$$\text{if } d_{0j} \le L_{\max}$$

$$x_{k,s-1} + x_{j,s} \leq 1, \qquad \forall k, j, s \text{ (if } s \text{ is even)} \qquad (13)$$

$$\text{if } d_{kj} \leq L$$

$$if\ d_{0k} + d_{kj} \geq L_{\max}$$

$$eff_{ik} = -49.74 + 5.897 \ln(\cos t_{ik}), \qquad \forall i, k \qquad (14)$$

$$x_{ks}, u_{kt}, scen_n \in \{0,1\} \qquad \forall k, s, n, t \qquad (15)$$

$$c_i, \cos t_{ik}, eff_{ik} \geq 0 \qquad \forall i, k \qquad (16)$$

The city by city allocated campaign budgets (y_{ik}) are removed from the original model so total budget is allocated only to cities that receive a campaign type. As a result, all the budget is being used in this new model. The decision variable used in the effect rate function is $cost_{ik}$ now because in this model each city receives a budget that is equal to the cost of campaign in that city. The rest of the model is same with the model in Chapter 3.2.

6.2. Base Model with Knapsack Approach

In this case study, the budget level for the base scenario is set at 55,000,000₺ because almost every penny of the budget could be spent in contrast to the former model. Also, gradual meeting and advertisement costs are implemented in this base model and all the parameters except the total budget remains the same with the previous base model in order to facilitate the comparison of the results.

Table 6.1 indicates that this approach significantly increases the total number of votes gained when compared to Table 5.3 because the budget is being allocated only to the cities that campaigns are run or in other words, all of the budget is spent. Since budget has a positive effect on the effect rate function, total number of votes gained has increased approximately by 4,000,000 with respect to the results of Table 5.3.

Table 6.1: Knapsack Approach Results – Base Model

Total Votes Gained	10,622,800	
Total Traveling Distance	14, 586 km	
Optimal Budget Allocation	60% Advertisement	40% Meeting
# of Cities to Run Each Campaign	Meetings in 24 Cities	Advertisement in 41 cities

6.3. Knapsack Approach with Weights

In this scenario, the new approach is added weights as in section 5.2.5. Briefly, the meeting campaigns' effect rate function is assigned a weight of 1.43 and the results are analyzed in Table 6.2.

Table 6.2: Knapsack Approach with Weight's Results – Base Model

Total Votes Gained	18,850,400	
Total Traveling Distance	24, 962 km	
Optimal Budget Allocation	40% Advertisement	60% Meeting
# of Cities to Run Each Campaign	Meetings in 40 Cities	Advertisement in 23 cities

Results in Table 6.2 imply that stronger meeting effects called for larger investments in meetings. The total meeting budget has increased to 33,000,000₺ from 22,000,000₺, therefore more cities are planned to be visited. Due to a decrease in the advertisement budget, the number of locations with advertisement decreased but this was compensated by the boosted meeting effects and ultimately, the objective functions' value has increased.

6.4. Knapsack Approach with Weights and Relaxed Distance Constraints

As the reasons stated in the previous chapter, the distance constraints will be relaxed again to analyze the effect of increased mobility in the campaign plan. So, the maximum distance between two consecutive cities that are visited in the same day (*L*) is set at 500 km and the maximum distance allowed before arriving to the second city in one day (*Lmax*) is set at 1,000 km.

Results in Table 6.3 demonstrates that the total number of votes gained has slightly increased compared to 400 km and 800 km for *L* and *Lmax* scenario. This increase stemmed from the longer distances politician could take. Although the number of cities visited remains the same, by allowing longer distances, further cities with higher vote potentials were chosen instead of lower potential but closer cities. For example, according to the previous scenario's results, a mass meeting was not planned in Erzurum as a consequence of its distance from Ankara, but in this scenario Erzurum could be selected and its' relatively high vote yield has increased the total number of votes. Likewise, Erzincan was given a mass meeting in the previous scenario, but in this scenario model was able to select higher potential cities even though they are further away.

Table 6.3: Knapsack Approach Considering Weights and Relaxed Distance Limits

Total Votes Gained	19,137,400	
Total Traveling Distance	26, 379 km	
Optimal Budget Allocation	40% Advertisement	60% Meeting
# of Cities to Run Each Campaign	Meetings in 40 Cities	Advertisement in 23 cities

6.5. Budget Analysis on Knapsack Approach

In this section, we present the results of the budget analysis on the knapsack approach (Section 6.3). That is, different budgets of 35,000,000₺, 45,000,000₺, 65,000,000₺ and 75,000,000₺ will be assigned to the scenario with weighted knapsack approach and relaxed distance constraints.

6.5.1. Total Budget of 35,000,000₺

In order to observe the effects of different budget levels, we both decrease and increase 55,000,000₺ as the total budget. 35,000,000₺ is chosen as the value to start with. Table 6.4 summarizes the results of this scenario.

Because of the weight assigned to the meetings, model decides to invest a large part of the budget in meeting campaigns (80%). The decrease in the total budget has resulted in a lower advertisement budget. According to the optimal results, 28,000,000₺ is spent on the mass meetings and 7,000,000₺ is spent on the advertisements. Since the meeting budget has not significantly changed, only seven less cities are given mass meetings compared to the previous scenario in

section 6.3. Therefore, the decrease in the objective function's value is mainly due to advertising in fewer locations as the number of towns with advertisement is 6 cities compared to 25 cities in scenario 6.3.

Table 6.4: Knapsack Approach Results – Budget: 35,000,000₺

Total Votes Gained	16,362,800	
Total Traveling Distance	21, 343 km	
Optimal Budget Allocation	20% Advertisement	80% Meeting
# of Cities to Run Each Campaign	Meetings in 33 Cities	Advertisement in 6 cities

6.5.2. Total Budget of 45,000,000₺

In the base model, we have 21 time periods and in each time period at most 2 cities can be visited: this means no matter how large the meeting budget is, model can at most select 42 cities for meetings. Moreover, the model has also distance constraints which implies that even if the model had 25 time periods to visit 50 cities, some of the cities may not have been given mass meetings because of these distance constraints.

According to the scenario in section 6.3, the best yielding route consists of 40 cities with a budget of 33,000,000₺. The last time period did not have any meetings because no more cities are available to satisfy the distance constraints. As long as the same weight is assigned to the meeting's effect rate function, model will always tend to complete the same route with route in 6.3 given the available budget. In other words, for this weighted scenario, the priority is always keeping the

meeting budget closest to 33,000,000₺ and visit those 40 cities because it brings the most votes and investing more in meetings is meaningless because of the distance constraints.

Table 6.5 shows the results for a total budget of 45,000,000₺. In this scenario, 31,500,000₺ of the 45,000,000₺ is allocated for mass meetings because this amount is the closest amount possible to 33,000,000₺ among the possible budget allocations shown in Table 4.4. The rest of the budget is allocated for advertisements in 13 cities. So, the increase in total votes is as a result of the increase in cities that the campaign was run.

Table 6.5: Knapsack Approach Results – Budget: 45,000,000₺

Total Votes Gained	18,016,600	
Total Traveling Distance	24, 411 km	
Optimal Budget Allocation	30% Advertisement	70% Meeting
# of Cities to Run Each Campaign	Meetings in 39 Cities	Advertisement in 13 cities

A budget analysis on 55,000,000₺ is not done separately here since it is already done in section 6.3.

6.5.3. Total Budget of 65,000,000₺

Results of this analysis (Table 6.6) imply that 39,000,000₺ is allocated for meetings and the rest (26,000,000₺) is spent on advertisement campaigns. As the reasons stated in 6.1.2, meeting budget is kept closest to 33,000,000₺ possible according to Table 4.4. Because of the distance limits, even though some budget is still available, model stopped visiting after the 40th city. Therefore, the total traveling distance remains the same with section 6.3 but the total number of

votes gained at the end of the campaign period has slightly increased to 19,524,300 because 6 more cities are given advertisements.

Table 6.6: Knapsack Approach Results – Budget: 65,000,000₺

Total Votes Gained	19,524,300	
Total Traveling Distance	26, 379 km	
Optimal Budget Allocation	40% Advertisement	60% Meeting
# of Cities to Run Each Campaign	Meetings in 40 Cities	Advertisement in 29 cities

6.5.4. Total Budget of 75,000,000₺

Results in Table 6.7 indicate that only 40% of the budget (30,000,000) is allocated for meetings. This may seem interesting since this scenario is the first among the other weighted scenarios that meetings received a lower portion from the total budget. In all other weighted scenarios, meetings took the higher portion of the budget since the vote yield was higher. But after a certain level on total budget, meeting effects reach a peak level and investing more becomes fruitless which is also the situation experienced in this scenario. As being said before, after 33,000,000₺ meeting's budget, total distance or number of visited cities does not change due to the distance criteria. So, model selects the budget allocation option that gives the closest budget to 33,000,000₺ for meeting expenses and spends the rest in advertisements. That is why, for 75,000,000₺ total budget, the best allocation would allocate 40% to meetings and 60% to advertisements as 75,000,000₺*0.4 equals 30,000,000₺.

Results also indicate that 10,000,000₺ budget increase does not create a huge difference in the objective function's value compared to Table 6.7 because of two reasons: the number of visited cities remains the same with scenario 6.4.3 and newly selected cities to give advertisements are all lightly populated cities which ultimately do not create a huge effect.

Table 6.7: Knapsack Approach Results – Budget: 75,000,000₺

Total Votes Gained	19,935,100	
Total Traveling Distance	23,702 km	
Optimal Budget Allocation	60% Advertisement	40% Meeting
# of Cities to Run Each Campaign	Meetings in 37 Cities	Advertisement 50 cities

CHAPTER 7

CONCLUSION

In this study, optimization of resource allocation and routing problem in political campaigns were combined for the first time in the literature to put forth a realistic campaign plan which can be adapted by any country and any political party as long as a proportional representation system is used as the electoral system.

The model was applied to two problems: one with 21 cities and one with 50 cities. The solution that the model provided indicated that some of the cities were selected for campaign appearances or advertisements or both. Besides, for the mass meetings, the model also provided the route for the campaign period. The results enabled the comparison between cities in terms of their importance for the political parties.

A detailed sensitivity analysis was made to observe the effect of different parameters in the model like nonstandard campaign costs and budgets, enabled partial advertisements and different effectiveness rates of the campaign types. At each section, a new assumption was built in to observe the cumulative effect. At the end of each analysis, model's priorities showed what direction the model needed to be improved. For instance, at the end of 5.2.2, it was realized that assuming nonstandard campaign costs necessitated nonstandard campaign budgets so in the next section these two assumptions were combined. Also, facilitating partial advertisement was a result of the same process.

In chapter 6, the model was reconsidered with no city by city assigned budgets for campaign types. The reason that caused this assumption is, in one of the scenarios, Istanbul which

is the most dominant city in Turkey, was not given a mass meeting because the budget allocated for Istanbul was less than the cost. Since gathering the biggest crowd ever in Istanbul was every politician's dream, the city by city allocated budgets were removed to improve the model.

One remarkable result is, mass meeting's effect does not increase continuously as the budget increases because of the limited campaign period and traveling distance restrictions. So, after allocating a certain amount for meetings, it is decided to spend the rest in advertisements.

Finally, this model will help the campaign planners how to solve the overwhelming campaign planning problem as it considers many aspects of the campaign as a whole and proposes integrated solutions.

CHAPTER 8

FUTURE RESEARCH

In this section, potential research directions are discussed. The results in the previous chapters indicate that the developed mathematical model is applicable to large case studies but there could still be some improvements in the model to obtain more accurate results.

The cost data used in the model is an estimation, although these estimations are done consistently with cities' populations; real costs of campaigns in each city can be incorporated in the model to get more realistic results. The validity of the model is tested with different scenarios but utilizing more accurate data will provide better insights.

In this study, a nonlinear effect rate function is utilized to convert the amount spent in each city to number of votes. This function can be analyzed in more details since it is derived from the limited data given in a prior study in the field and for both campaign types, the same function is used. If more data can be found to create different functions, the model can incorporate the new effect rate function. Also, this function can be linearized to get a higher correlation coefficient for better insights.

In this thesis, we considered two different networks which have 20 and 50 cities but number of cities could be increased to 81 (total number of cities in Turkey) to propose a real campaign plan. Another improvement can be a model that decides the budget allocation options without restricting to any number of pre-determined options, i.e., budget allocation variable can be a continuous variable. Also, the historical data from different parties' political campaigns could be

gathered to use as a reference to measure the validity and applicability of the results provided by the model.

Different cases are investigated in the sensitivity analysis section: some of the results have shown that either a minimal number or none of the cities was selected for meetings or advertisements. But in real life, political parties would not agree on this because some cities are considered extremely important to run the campaign. Therefore, a minimum budget for some of the cities for advertisements or meetings can be included in the mathematical model.

Timing of the campaigns is an important factor. Sometimes politicians save the most sensational activities or arguments of the campaign for the last which is how the recency effect is exploited in political campaigns. Nickerson (2007) stated that campaign related activities done in a week before the elections stimulated voter turnout. So, in order to investigate the effects of timing in this work, the effect rate of the meetings that are hold towards the end of the campaign period may be increased to see how the model responds.

The above listed future research directions would help improving the research presented in this thesis.

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APPENDIX

APPENDIX A

CITY DISTANCE MATRIX FOR 50 CITY CASE STUDY

ORDER OF CITY	CITIES	ADANA	ADIYAMAN	ANKARA	ANTALYA	BALIKESİR	BİNGÖL	BURSA	ÇORUM	DENİZLİ	DİYARBAKIR	EDİRNE	ELAZIĞ	ERZİNCAN	ERZURUM	ESKİŞEHİR	GAZİANTEP	GİRESUN	HAKKARİ	HATAY	MERSİN	İSTANBUL	İZMİR	KARS	KASTAMONU	KAYSERİ	KIRKLARELİ	KOCAELİ	KONYA	KÜTAHYA	MALATYA	MANİSA	KAHRAMANMARAŞ	MARDİN	MUĞLA	NİĞDE	ORDU	RİZE	SAKARYA	SAMSUN	SİNOP	SİVAS	TEKİRDAĞ	TRABZON	ŞANLIURFA	VAN	YOZGAT	ZONGULDAK	KIRIKKALE	ŞIRNAK	YALOVA
1	ADANA	0	335	489	546	900	634	838	578	759	525	1176	494	671	810	687	212	721	909	191	69	946	899	1014	680	332	1157	835	355	672	396	883	192	537	857	204	710	920	798	722	847	423	1078	845	349	895	486	764	473	720	898
2	ADIYAMAN	335	0	746	881	1235	351	1117	699	1094	207	1432	285	550	531	966	150	713	671	317	404	1202	1234	735	882	427	1413	1091	690	1007	187	1218	163	298	1192	539	729	859	1054	753	893	415	1334	784	112	577	624	1017	675	482	1156
3	ANKARA	489	746	0	544	535	893	384	241	475	904	683	753	685	875	233	659	608	1356	680	482	453	579	1076	237	319	664	342	258	311	655	563	583	984	620	347	564	819	305	413	413	439	585	744	796	1220	215	268	75	1167	407
4	ANTALYA	546	881	544	0	509	1180	541	733	220	1071	913	1040	1058	1248	424	758	1100	1455	737	477	718	444	1449	774	618	929	607	322	364	942	428	738	1083	311	544	1056	1307	570	905	950	810	850	1232	895	1441	684	752	567	1266	605
5	BALIKESİR	900	1235	535	509	0	1415	151	777	289	1425	407	1275	1201	1391	302	1112	1093	1809	1091	893	390	176	1592	670	841	424	279	551	228	1177	141	1092	1437	395	787	1049	1304	311	898	846	982	379	1229	1249	1742	758	493	618	1620	216
6	BİNGÖL	634	351	893	1180	1415	0	1264	731	1282	144	1541	142	273	180	1113	457	546	503	616	703	1311	1422	384	893	574	1522	1200	878	1191	238	1406	462	239	1427	702	590	559	1163	719	864	468	1443	484	320	331	692	1109	822	379	1265
7	BURSA	838	1117	384	541	151	1264	0	626	438	1275	419	1124	1050	1240	151	1030	942	1727	1029	831	243	325	1441	519	690	436	132	489	177	1026	290	954	1355	544	716	898	1153	160	747	695	831	375	1078	1167	1591	607	342	467	1538	69
8	ÇORUM	578	699	241	733	777	731	626	0	724	763	844	610	458	648	482	621	367	1230	719	571	614	828	849	196	281	825	503	419	560	530	812	545	858	869	374	323	578	466	172	269	284	746	503	758	1058	108	412	166	1045	568
9	DENİZLİ	759	1094	475	220	289	1282	438	724	0	1284	693	1142	1148	1338	355	971	1091	1668	950	697	639	224	1539	723	708	710	528	410	285	1044	208	951	1296	145	646	1047	1302	491	896	899	900	665	1227	1108	1609	698	673	558	1479	503
10	DİYARBAKIR	525	207	904	1071	1425	144	1275	763	1284	0	1590	153	408	324	1124	313	690	471	507	594	1360	1424	528	953	585	1571	1249	880	1197	249	1408	370	95	1382	713	733	703	1212	817	957	479	1492	628	176	370	703	1169	833	282	1314
11	EDİRNE	1176	1432	683	913	407	1541	419	844	693	1590	0	1439	1268	1458	554	1345	1160	2040	1367	1169	230	536	1659	737	1005	62	341	892	584	1341	539	1269	1670	759	1034	1116	1371	378	965	913	1122	140	1296	1482	1868	901	560	761	1853	406
12	ELAZIĞ	494	285	753	1040	1275	142	1124	610	1142	153	1439	0	267	318	973	342	563	624	476	563	1209	1282	522	800	434	1420	1098	738	1051	98	1266	322	248	1287	562	592	576	1061	664	804	326	1341	501	319	469	550	1016	682	435	1163
13	ERZİNCAN	671	550	685	1058	1201	273	1050	458	1148	408	1268	267	0	190	926	607	296	772	741	740	1038	1272	391	620	442	1249	927	744	1004	363	1256	578	503	1293	568	325	309	890	446	591	248	1170	234	574	600	472	836	610	652	992
14	ERZURUM	810	531	875	1248	1391	180	1240	648	1338	324	1458	318	190	0	1116	637	366	616	792	879	1228	1462	204	810	632	1439	1117	934	1194	414	1446	638	419	1483	758	410	379	1080	561	716	438	1360	304	500	414	662	1026	800	541	1182
15	ESKİŞEHİR	687	966	233	424	302	1113	151	482	355	1124	554	973	926	1116	0	879	849	1576	878	680	324	411	1317	481	539	535	213	338	78	875	395	803	1204	500	565	805	1060	176	654	657	680	456	985	1016	1440	456	358	316	1387	211
16	GAZİANTEP	212	150	659	758	1112	457	1030	621	971	313	1345	342	607	637	879	0	704	697	194	281	1115	1111	841	795	340	1326	1004	567	884	244	1095	76	325	1069	416	693	903	967	705	845	406	1247	828	137	683	537	930	588	508	1069
17	GİRESUN	721	713	608	1100	1093	546	942	367	1091	690	1160	563	296	366	849	704	0	979	802	790	930	1195	567	485	492	1141	819	786	927	544	1179	628	785	1236	618	44	211	782	195	350	298	1062	136	797	777	467	701	533	904	884
18	HAKKARİ	909	671	1356	1455	1809	503	1727	1230	1668	471	2040	624	772	616	1576	697	979	0	891	978	1810	1808	567	1392	1037	2021	1699	1264	1581	720	1792	773	398	1766	1113	1023	924	1662	1174	1329	950	1942	917	560	202	1174	1608	1285	189	1764
19	HATAY	191	317	680	737	1091	616	1029	719	950	507	1367	476	741	792	878	194	802	891	0	260	1137	1090	996	871	438	1348	1026	546	863	378	1074	174	519	1048	395	791	1001	989	803	943	504	1269	926	331	877	635	955	664	702	1089
20	MERSİN	69	404	482	477	893	703	831	571	697	594	1169	563	740	879	680	281	790	978	260	0	939	892	1083	673	325	1150	828	348	665	465	876	261	606	788	197	779	989	791	743	840	492	1071	914	418	964	479	757	466	789	891
21	İSTANBUL	946	1202	453	718	390	1311	243	614	639	1360	230	1209	1038	1228	324	1115	930	1810	1137	939	0	564	1429	507	775	211	111	662	354	1111	529	1039	1440	783	804	886	1141	148	735	683	892	132	1066	1252	1638	671	330	531	1623	176
22	İZMİR	899	1234	579	444	176	1422	325	828	224	1424	536	1282	1272	1462	411	1111	1195	1808	1090	892	564	0	1663	827	848	553	453	550	333	1184	35	1091	1436	225	786	1151	1406	485	1000	1003	1026	508	1331	1248	1749	802	667	662	1619	390
23	KARS	1014	735	1076	1449	1592	384	1441	849	1539	528	1659	522	391	204	1317	841	567	567	996	1083	1429	1663	0	1011	833	1640	1318	1135	1395	618	1647	842	623	1684	959	611	363	1281	762	917	639	1561	438	704	365	863	1227	1001	625	1383
24	KASTAMONU	680	882	237	774	670	893	519	196	723	953	737	800	620	810	481	795	485	1392	871	673	507	827	1011	0	455	718	396	460	559	720	809	719	1048	868	494	441	696	359	290	183	474	639	621	932	1220	300	216	211	1235	461
25	KAYSERİ	332	427	319	618	841	574	690	281	708	585	1005	434	442	632	539	340	492	1037	438	325	775	848	833	455	0	986	664	304	617	336	832	264	665	853	128	481	691	627	453	550	194	907	616	477	901	197	590	248	848	729

APPENDIX A (continued)

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